

A Layered Framework for Compact-Resolvent Spectral Encodings

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Abstract

Spectral generating functions are often discussed as though compactness of an underlying carrier, discreteness of an operator spectrum, trace-class heat evolution, and geometric realizability were interchangeable. They are not. This expository framework separates these assertions into typed layers. The operator layer consists of a nonnegative self-adjoint operator with compact resolvent; heat-admissibility is imposed independently. Its spectral counting measure

$$\mu_L = \sum_j m_j \delta_{\nu_j}$$

produces the heat trace and its complex-time deformation as Laplace transforms

$$Z_L(z) = \int_{[0,\infty)} e^{-z\lambda} d\mu_L(\lambda), \quad \Re z > 0.$$

We prove the elementary holomorphy, differentiation, domination, and exact imaginary-periodicity criteria for this transform; correct the factorization criterion for quantities recoverable through a readout; and distinguish formal, analytic, abstract diagonal, elliptic-geometric, equivariant, and physical realization levels. Circle and flat-torus Laplacians provide normalized theta examples. Two counterexamples show that a compact carrier need not produce discrete spectrum and that compact resolvent need not produce a trace-class heat semigroup. A rapidly growing abstract multiplicity model then exhibits the gap between diagonal operator realizability and finite-dimensional elliptic geometry. No new theorem in spectral geometry is claimed; the contribution is a rigorous interface and failure-mode taxonomy assembled from standard operator and heat-kernel theory.

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Status and scope. This is an expository framework article. Except for elementary interface propositions proved here for completeness, its analytic inputs are standard results in functional analysis, elliptic theory, harmonic analysis, and compact group representation theory. It does not propose a new spectral theory, a physical compactification model, or an inverse spectral theorem.

1 Introduction

The phrase “compact spectral geometry” can conceal several logically independent assertions. A carrier may be compact while a chosen operator has continuous spectrum. A nonnegative self-adjoint operator may have compact resolvent while its heat semigroup fails to be trace class for small positive times. A formal power series may admit an abstract diagonal operator realization while being incompatible with the Weyl and heat asymptotics of every Laplace-type operator on a closed finite-dimensional manifold. Finally, a representation label attached to an eigenspace is algebraic data, not by itself a physical particle sector.

The purpose of this article is to make these boundaries explicit. The basic chain is

operator hypotheses $\longrightarrow$ spectral measure $\longrightarrow$ analytic encoding $\longrightarrow$ declared readout.
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Each arrow has hypotheses. None may be replaced by an analogy.

1.1 The layered contract

We use the following hierarchy.

1. **Formal layer.** An expression $\sum_i a_i q^{\lambda_i}$ is symbolic data until the index set, levels, coefficients, convergence regime, and branch convention are specified.
2. **Abstract operator layer.** A pair $(\mathcal{H}, L)$, with L nonnegative self-adjoint and compact-resolvent, supplies a discrete spectral counting measure and finite spectral windows.
3. **Heat layer.** The extra condition $e^{-tL} \in \mathcal{S}_1$ for every $t > 0$ supplies a heat trace and a holomorphic complex-time trace.
4. **Geometric layer.** A closed manifold, Hermitian bundle, and Laplace-type operator supply compact resolvent, heat-admissibility, Weyl growth, and a local small-time expansion.
5. **Equivariant/readout layer.** A commuting compact-group action and a specified map identify symmetry labels and exactly which quantities descend through the map.
6. **Physical layer.** An action, units, observables, parameter identification, and empirical tests are additional data. They are outside the present framework.

The separation between the abstract and geometric layers is essential. Every suitable discrete multiplicity list has a diagonal realization, whereas local elliptic realizations obey strong asymptotic constraints.

1.2 Relation to standard frameworks

Compact-resolvent spectral theory, trace ideals, heat kernels, and elliptic heat asymptotics are classical; see [13, 9, 15, 5, 10, 8, 14]. The circle and torus formulas are direct instances of Fourier analysis and Poisson summation [16, 11]. The compact-group statements are standard consequences of the spectral theorem and complete reducibility [6, 7].

The datum used below is deliberately weaker than a spectral triple. It contains no represented algebra, bounded-commutator condition, real structure, grading, or dimension axiom, and therefore does not inherit conclusions of noncommutative geometry [4]. It also contains no spectral-action functional [2]. The present contribution is organizational: it gives a minimal typed interface, records the exact descent criterion for readouts, and isolates counterexamples that invalidate common shortcuts.

2 Compact resolvent and heat-admissibility

2.1 Operator data

Definition 2.1 (Abstract compact-resolvent datum). An abstract compact-resolvent spectral datum is a pair $(\mathcal{H}, L)$ such that $\mathcal{H}$ is a complex separable Hilbert space and

$$L : \text{Dom}(L) \subset \mathcal{H} \longrightarrow \mathcal{H}$$

is densely defined, nonnegative, self-adjoint, and has compact resolvent. Because $L \geq 0$, it is enough to require $(L + \mathbf{I})^{-1}$ to be compact.

This definition does not mention a carrier. A carrier becomes relevant only after a concrete realization of $\mathcal{H}$ and L has been specified.

Theorem 2.2 (Standard compact-resolvent spectral theorem). *Let $(\mathcal{H}, L)$ be an abstract compact-resolvent datum.*

1. *If $\dim \mathcal{H} < \infty$, then L has a finite orthonormal eigenbasis.*

2. If $\dim \mathcal{H} = \infty$, then there is an orthonormal basis $(e_j)_{j \geq 0}$ and a sequence

$$0 \leq \lambda_0 \leq \lambda_1 \leq \cdots, \quad \lambda_j \longrightarrow \infty,$$

such that $Le_j = \lambda_j e_j$. Eigenvalues are repeated according to finite multiplicity.

3. For every bounded Borel set $W \subset \mathbb{R}$, the spectral projection $\mathbf{1}_W(L)$ has finite rank.

Proof. The compact self-adjoint operator $R = (L + \mathbf{I})^{-1}$ has an orthonormal eigenbasis for its nonzero spectrum together with a basis of $\ker R$. Here R is injective, hence $\ker R = \{0\}$, so the nonzero eigenvectors are complete. In infinite dimension, the nonzero eigenvalues of R have finite multiplicity and can accumulate only at zero. If $Re = \rho e$, then $e \in \text{ran } R = \text{Dom}(L)$ and $Le = (\rho^{-1} - 1)e$. This transfers the basis to L , gives finite multiplicities, and forces $\lambda_j \rightarrow \infty$. The finite-dimensional alternative must be stated separately. In either case, a bounded set meets only finitely many eigenvalues counted with multiplicity, so its spectral projection has finite rank. See [12, Chs. VI–VII] or [9, Ch. III]. $\square$

Definition 2.3 (Spectral window). For a bounded Borel set $W \subset [0, \infty)$, the spectral window is

$$\mathcal{H}_W := \text{ran } \mathbf{1}_W(L).$$

By Theorem 2.2, $\mathcal{H}_W$ is finite-dimensional.

2.2 Compact carrier is not an operator hypothesis

Example 2.4 (Compact carrier with continuous spectrum). Let $K = S^1$, $\mathcal{H} = L^2(S^1)$, and

$$(Lf)(\theta) = (2 + \cos \theta)f(\theta).$$

Then K is compact and L is bounded, nonnegative, and self-adjoint, but

$$\text{Spec}(L) = [1, 3],$$

the essential range of $2 + \cos \theta$. Moreover, $(L + \mathbf{I})^{-1}$ is multiplication by $(3 + \cos \theta)^{-1}$, which is not compact on $L^2(S^1)$. Indeed, this multiplier has the bounded inverse given by multiplication by $3 + \cos \theta$; if it were compact, the identity on the infinite-dimensional space $L^2(S^1)$ would be compact. Thus compactness of the carrier alone gives neither compact resolvent nor discrete spectrum.

Remark 2.5. Compactness becomes effective only through an analytic theorem. For a Laplace-type operator on a closed manifold, elliptic regularity maps the resolvent into a higher Sobolev space and Rellich compactness returns it compactly to L^2 . The conclusion comes from the operator and its domain, not from the topology of the carrier in isolation.

2.3 Heat-admissibility is an independent condition

Definition 2.6 (Heat-admissibility). A nonnegative self-adjoint operator L is heat-admissible if

$$e^{-tL} \in \mathcal{S}_1(\mathcal{H}) \quad \text{for every } t > 0,$$

where $\mathcal{S}_1$ is the trace-class ideal. Its heat trace is

$$Z_L(t) := \text{Tr}(e^{-tL}).$$

Remark 2.7. This is a naming convention for the present layer structure, not a new summability notion. When $L = D^2$, the same condition is commonly called θ -summability in spectral-triple terminology [4, 15].

Example 2.8 (Compact resolvent without heat-admissibility). On $\ell^2(\mathbb{N})$, define

$$Le_n = \log(n+1)e_n, \quad \text{Dom}(L) = \left\{ x : \sum_{n \geq 1} \log^2(n+1) |x_n|^2 < \infty \right\}.$$

Then $L \geq 0$ is self-adjoint and $(L + I)^{-1}$ is diagonal with entries tending to zero, hence compact. Self-adjointness follows from the maximal-domain construction for a real diagonal operator. However,

$$\text{Tr}(e^{-tL}) = \sum_{n \geq 1} (n+1)^{-t},$$

which diverges for $0 < t \leq 1$. Compact resolvent therefore does not imply heat-admissibility.

Remark 2.9 (Growth criterion). If λ_j are repeated with multiplicity, heat-admissibility is exactly

$$\sum_j e^{-t\lambda_j} < \infty \quad \text{for every } t > 0.$$

It is a quantitative restriction on spectral growth, not merely a discreteness condition.

3 Spectral measures and complex-time encodings

3.1 Counting measure and heat trace

Let $\nu_0 < \nu_1 < \dots$ denote the distinct eigenvalues of a compact-resolvent operator and let m_j be their multiplicities. The list is finite in finite dimension; otherwise it is an infinite sequence with $\nu_j \rightarrow \infty$.

Definition 3.1 (Spectral counting measure). The spectral counting measure and counting function are

$$\mu_L := \sum_j m_j \delta_{\nu_j}, \quad N_L(\Lambda) := \mu_L([0, \Lambda]).$$

The measure μ_L is locally finite. If L is heat-admissible, then

$$Z_L(t) = \int_{[0, \infty)} e^{-t\lambda} d\mu_L(\lambda) = \sum_j m_j e^{-t\nu_j}.$$

Thus the heat trace is the Laplace transform of a positive discrete measure. This formulation keeps levels and multiplicities distinct and avoids assigning spectral meaning to an undeclared coefficient sequence.

3.2 Complex time

Definition 3.2 (Complex-time spectral trace). For a heat-admissible L and $z \in \mathbb{C}$ with $\Re z > 0$, define

$$Z_L(z) := \text{Tr}(e^{-zL}) = \int_{[0, \infty)} e^{-z\lambda} d\mu_L(\lambda).$$

Writing $z = t - i\phi$, with $t > 0$ and $\phi \in \mathbb{R}$, gives

$$Z_L(t - i\phi) = \sum_j m_j e^{-t\nu_j} e^{i\phi\nu_j}.$$

The phase parameter is initially real, not an element of $\mathbb{R}/2\pi\mathbb{Z}$.

Proposition 3.3 (Holomorphic trace and derivative bounds). *Let L be heat-admissible. Then:*

1. e^{-zL} is trace class for every $\Re z > 0$, and

$$\|e^{-zL}\|_1 = Z_L(\Re z).$$

2. Z_L is holomorphic on the right half-plane and, for every $r \in \mathbb{N}_0$,

$$Z_L^{(r)}(z) = (-1)^r \int \lambda^r e^{-z\lambda} d\mu_L(\lambda).$$

3. The domination estimate

$$|Z_L(z)| \leq Z_L(\Re z)$$

holds.

Proof. Functional calculus gives

$$e^{-zL} = e^{-(\Re z)L} e^{-i(\Im z)L}.$$

The second factor is unitary and commutes with the positive trace-class first factor, giving the trace norm identity. On every compact subset of $\{\Re z > 0\}$, there is $a > 0$ with $\Re z \geq a$. The defining series is dominated by $\sum_j m_j e^{-a\nu_j} < \infty$, so it converges locally uniformly. For derivatives, use

$$\lambda^r e^{-a\lambda} \leq C_{r,a} e^{-a\lambda/2}$$

and heat-admissibility at $a/2$. Termwise differentiation is therefore valid. The final estimate follows from the triangle inequality. $\square$

Proposition 3.4 (Exact imaginary-periodicity criterion). *Let L be heat-admissible and let $T > 0$. The following are equivalent:*

1. $Z_L(z + iT) = Z_L(z)$ for all $\Re z > 0$;
2. $T\nu_j \in 2\pi\mathbb{Z}$ for every eigenvalue ν_j .

In particular, 2π -periodicity in ϕ is automatic only when all levels are integers.

Proof. The second condition immediately implies the first. Conversely, evaluate the assumed identity at real $z = t > 0$:

$$\sum_j m_j (e^{-iT\nu_j} - 1) e^{-t\nu_j} = 0.$$

Order the distinct eigenvalues increasingly and set $c_j = m_j(e^{-iT\nu_j} - 1)$. For any fixed $t_0 > 0$ and $t \geq t_0$,

$$|c_j| e^{-t(\nu_j - \nu_0)} \leq 2e^{t_0\nu_0} m_j e^{-t_0\nu_j},$$

and the majorant is summable by heat-admissibility. Multiplying the identity by $e^{t\nu_0}$ and applying dominated convergence as $t \rightarrow \infty$ therefore gives $c_0 = 0$. Remove that term and apply the same argument to the tail beginning at ν_1 , then iterate. Hence $e^{-iT\nu_j} = 1$, equivalently $T\nu_j \in 2\pi\mathbb{Z}$, for every j . $\square$

Remark 3.5 (Safe q -notation). If the levels are nonnegative integers, one may set $q = e^{-z}$, $|q| < 1$, and write $Z_L(z) = \sum_n m_n q^n$. For general real levels, q^λ requires a choice of logarithm. The invariant notation is $e^{-z\lambda}$.

3.3 Mellin transform and the positive spectral zeta function

Definition 3.6. Let $h_L = \dim \ker L$. The positive spectral zeta function is

$$\zeta_L(s) := \int_{(0,\infty)} \lambda^{-s} d\mu_L(\lambda) = \sum_{\nu_j > 0} m_j \nu_j^{-s}$$

in its half-plane of absolute convergence.

Proposition 3.7 (Mellin identity with explicit hypothesis). *Let L be heat-admissible, let $s \in \mathbb{C}$, let $\sigma = \Re s > 0$, and assume $\sum_{\nu_j > 0} m_j \nu_j^{-\sigma} < \infty$. Then*

$$\zeta_L(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} (Z_L(t) - h_L) dt.$$

Proof. For $\lambda > 0$,

$$\lambda^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-t\lambda} dt.$$

Moreover, Tonelli's theorem gives the exact absolute-integrability test

$$\int_0^\infty \sum_{\nu_j > 0} m_j t^{\sigma-1} e^{-t\nu_j} dt = \Gamma(\sigma) \sum_{\nu_j > 0} m_j \nu_j^{-\sigma} < \infty.$$

Fubini's theorem therefore gives the identity. No analytic continuation is asserted. $\square$

4 A realization hierarchy

4.1 Formal and abstract realizations

Definition 4.1 (Level encoding). A level encoding is data (I, λ, a) , where I is countable, $\lambda : I \rightarrow [0, \infty)$, and $a : I \rightarrow \mathbb{C}$. It is *locally finite* if

$$\#\{i \in I : a_i \neq 0, \lambda_i \leq \Lambda\} < \infty \quad (\Lambda < \infty).$$

On an open set $D \subset \mathbb{C}$, it defines an analytic encoding if the series

$$F(z) = \sum_{i \in I} a_i e^{-z\lambda_i}$$

converges normally on D , meaning that for every compact $K \subset D$,

$$\sum_{i \in I} \sup_{z \in K} |a_i e^{-z\lambda_i}| < \infty.$$

Normal convergence makes F holomorphic. Under local finiteness, the grouped coefficient at a level ν is the finite sum

$$b_\nu := \sum_{\lambda_i = \nu} a_i.$$

The encoding can be an abstract heat trace only if every b_ν is a nonnegative integer, the support $\{\nu : b_\nu \neq 0\}$ is locally finite in $[0, \infty)$, and $\sum_\nu b_\nu e^{-t\nu} < \infty$ for every $t > 0$.

Proposition 4.2 (Canonical diagonal realization). *Let*

$$0 \leq \lambda_0 < \lambda_1 < \cdots, \quad \lambda_n \rightarrow \infty, \quad m_n \in \mathbb{N}_{>0}.$$

On

$$\mathcal{H} = \bigoplus_{n \geq 0} \mathbb{C}^{m_n}$$

define L to act by $\lambda_n \mathbf{I}$ on the n -th summand, with domain

$$\text{Dom}(L) = \left\{ x = (x_n) : \sum_{n \geq 0} \lambda_n^2 \|x_n\|^2 < \infty \right\}.$$

Then $L \geq 0$ is self-adjoint with compact resolvent. It is heat-admissible if and only if

$$\sum_{n \geq 0} m_n e^{-t\lambda_n} < \infty \quad \text{for every } t > 0.$$

The resulting pure-point operator is unique up to unitary equivalence among operators with exactly these eigenvalues and multiplicities. For a finite level list, the same statement holds on a finite-dimensional direct sum, with the condition $\lambda_n \rightarrow \infty$ omitted.

Proof. The displayed domain is the maximal domain of this real diagonal operator, so L is self-adjoint and nonnegative. The resolvent is block diagonal with block norm $(1 + \lambda_n)^{-1} \rightarrow 0$, hence it is the norm limit of finite-rank truncations. The trace-class criterion follows by summing the positive eigenvalues $m_n e^{-t\lambda_n}$ of e^{-tL} . Unitary equivalence follows by choosing unitary identifications between corresponding finite-dimensional eigenspaces. $\square$

Warning 4.3. Proposition 4.2 makes abstract realizability inexpensive. It does not produce a carrier, locality, differential expression, principal symbol, dimension, metric, boundary condition, or physical interpretation.

4.2 Elliptic geometric realization

Definition 4.4 (Closed Laplace-type realization). A closed Laplace-type realization consists of a closed smooth d -dimensional Riemannian manifold (M, g) , $d \geq 1$, a finite-rank Hermitian bundle $E \rightarrow M$, and the nonnegative self-adjoint realization on $L^2(M, E)$ of a second-order differential operator L whose principal symbol is

$$\sigma_2(L)(x, \xi) = |\xi|_g^2 \mathbf{I}_{E_x}.$$

Theorem 4.5 (Standard elliptic package). *For a closed Laplace-type realization:*

1. L has compact resolvent and is heat-admissible;
2. its counting function satisfies the Weyl law

$$N_L(\Lambda) \sim \frac{\text{rank}(E) \omega_d}{(2\pi)^d} \text{Vol}(M) \Lambda^{d/2}, \quad \Lambda \rightarrow \infty;$$

3. its heat trace has a complete small-time expansion

$$Z_L(t) \sim (4\pi t)^{-d/2} \sum_{k=0}^{\infty} A_k(L) t^k, \quad t \downarrow 0,$$

with $A_0(L) = \text{rank}(E) \text{Vol}(M)$ and higher coefficients determined by local geometric and bundle data.

Source. Compact resolvent follows from elliptic regularity and Rellich compactness. The heat-kernel construction, local expansion, and Weyl law are classical; see [10, 8, 5, 1, 3, 14]. This theorem is quoted as a standard package; it is not presented as a new result. $\square$

Remark 4.6. The displayed asymptotics are necessary tests for a closed Laplace-type realization. They are not sufficient: matching a finite or even an asymptotic list of coefficients does not construct a manifold and operator. Boundaries, singular spaces, noncompact carriers, nonlocal operators, and elliptic operators of order other than two have different theorem packages and must be declared separately.

Example 4.7 (Abstract heat trace excluded by elliptic asymptotics). Let $\lambda_n = n$ and $m_n = \lceil e^{\sqrt{n}} \rceil$ for $n \geq 1$. Proposition 4.2 gives a compact-resolvent operator. It is heat-admissible because $\sum_n e^{\sqrt{n} - tn} < \infty$ for every $t > 0$. For $0 < t \leq 1/4$, choose $n_t = \lfloor (4t^2)^{-1} \rfloor$. Then, for a constant $c > 0$,

$$Z_L(t) \geq m_{n_t} e^{-tn_t} \geq e^{c/t}$$

once t is sufficiently small. Hence $Z_L(t)$ grows faster than every power of t^{-1} . It cannot be the heat trace of a Laplace-type operator on any closed finite-dimensional manifold, because that would contradict Theorem 4.5.

Proof. For fixed $t > 0$, the inequality $\sqrt{n} \leq tn/2$ holds once $n \geq 4t^{-2}$; hence the tail of $\sum_n e^{\sqrt{n}-tn}$ is bounded by a geometric series $\sum_n e^{-tn/2}$. This proves heat-admissibility.

For $t \leq 1/4$, the number $x = (4t^2)^{-1}$ is at least 4, so $n_t \geq x/2 = (8t^2)^{-1}$, while $tn_t \leq 1/(4t)$. Therefore

$$\sqrt{n_t} - tn_t \geq \left(\frac{1}{2\sqrt{2}} - \frac{1}{4} \right) \frac{1}{t}.$$

The coefficient in parentheses is positive, which proves the bound. $\square$

4.3 Layer table

Layer	Required data	What is justified
Formal	I, λ, a , syntax	A symbolic level expression only.
Convergent analytic	Normal convergence on an open set	A holomorphic function on that set.
Abstract spectral	Integer multiplicities and a self-adjoint realization	A pure-point spectrum; compact resolvent if levels escape to infinity.
Heat	$e^{-tL} \in \mathcal{S}_1$ for all $t > 0$	Trace, spectral Laplace transform, complex-time encoding.
Elliptic geometric	M, E, L , domain and ellipticity	Weyl law, local heat expansion, geometric coefficients.
Equivariant	Strongly continuous unitary action commuting with L	Representation structure on finite eigenspaces.
Physical	Dynamics, units, observables, calibration and tests	Model-dependent physical statements only.

5 Circle and flat-torus models

We use the Fourier-transform convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx,$$

for which the lattice Poisson formula is $\sum_{n \in \mathbb{Z}^d} f(n) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k)$ on the Schwartz class.

5.1 Circle

Let $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ and equip it with normalized Haar measure $d\theta/(2\pi)$. For $R > 0$, define

$$L_R = -R^{-2} \frac{d^2}{d\theta^2}, \quad \text{Dom}(L_R) = H_{\text{per}}^2(S^1).$$

Proposition 5.1 (Circle spectrum and theta trace). *The modes $e_n(\theta) = e^{in\theta}$, $n \in \mathbb{Z}$, form an orthonormal eigenbasis with*

$$L_R e_n = \frac{n^2}{R^2} e_n.$$

Consequently,

$$Z_{L_R}(t) = \sum_{n \in \mathbb{Z}} e^{-tn^2/R^2} = R \sqrt{\frac{\pi}{t}} \sum_{k \in \mathbb{Z}} e^{-\pi^2 R^2 k^2/t}.$$

Proof. The spectral statement is Fourier theory on S^1 . Apply Poisson summation to $x \mapsto e^{-tx^2/R^2}$ to obtain the second expression. $\square$

The equality relates short and long heat scales, but it is a precise identity for this quadratic spectrum, not a universal duality principle.

5.2 Flat tori

Let $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ with normalized Haar measure, let A be a real symmetric positive-definite $d \times d$ matrix, and define

$$L_A = - \sum_{a,b=1}^d A_{ab} \partial_a \partial_b \quad \text{on } H^2(\mathbb{T}^d).$$

Proposition 5.2 (Flat-torus spectrum and Poisson transform). *The normalized Fourier modes*

$$e_n(\theta) = e^{in \cdot \theta}, \quad n \in \mathbb{Z}^d,$$

satisfy

$$L_A e_n = (n^\top A n) e_n.$$

The heat trace is

$$Z_A(t) = \sum_{n \in \mathbb{Z}^d} e^{-tn^\top A n},$$

and Poisson summation gives

$$Z_A(t) = \frac{\pi^{d/2}}{t^{d/2} \sqrt{\det A}} \sum_{k \in \mathbb{Z}^d} \exp \left(-\frac{\pi^2}{t} k^\top A^{-1} k \right).$$

Proof. Differentiating e_n gives the eigenvalue $n^\top A n$. Positive definiteness yields $n^\top A n \geq c|n|^2$, so the heat trace converges. The Fourier transform of $x \mapsto e^{-tx^\top A x}$ is

$$\xi \mapsto \frac{\pi^{d/2}}{t^{d/2} \sqrt{\det A}} \exp \left(-\frac{\pi^2}{t} \xi^\top A^{-1} \xi \right),$$

and the lattice Poisson formula proves the identity. $\square$

Remark 5.3 (Degeneracy). The multiplicity of λ is the finite representation number

$$r_A(\lambda) = \# \{ n \in \mathbb{Z}^d : n^\top A n = \lambda \}.$$

A statement about one Fourier label n must not be confused with a statement about the full eigenspace when several lattice vectors have the same quadratic value.

5.3 Tensor sums

Proposition 5.4 (Product heat trace). *Let L_i be a nonnegative heat-admissible self-adjoint operator on $\mathcal{H}_i$, $i = 1, 2$. On $\mathcal{H}_1 \widehat{\otimes} \mathcal{H}_2$, let L be the nonnegative self-adjoint operator associated with the closed quadratic form*

$$\text{Dom}(q) = \text{Dom}(L_1^{1/2} \otimes \text{I}) \cap \text{Dom}(\text{I} \otimes L_2^{1/2}), \quad q[u] = \left\| (L_1^{1/2} \otimes \text{I})u \right\|^2 + \left\| (\text{I} \otimes L_2^{1/2})u \right\|^2$$

for $u \in \text{Dom}(q)$. Equivalently, L is the self-adjoint tensor sum $L_1 \otimes \text{I} + \text{I} \otimes L_2$. Then L is heat-admissible and, for $\Re z > 0$,

$$e^{-zL} = e^{-zL_1} \otimes e^{-zL_2}, \quad Z_L(z) = Z_{L_1}(z) Z_{L_2}(z).$$

Proof. The spectral measures of the two tensor factors commute, and their joint functional calculus represents L by the function $(\lambda_1, \lambda_2) \mapsto \lambda_1 + \lambda_2$. This gives the exponential identity. For real $z = t > 0$, the tensor product of the two positive trace-class heat operators is trace class, proving heat-admissibility. The same trace-ideal fact for complex z , together with $\text{Tr}(A \otimes B) = \text{Tr}(A) \text{Tr}(B)$, gives the trace formula. $\square$

6 Readouts and equivariant labels

6.1 Exact factorization through a readout

Definition 6.1 (Readout and recoverability). Let $\Pi : X \rightarrow Y$ be a specified map and $Q : X \rightarrow Z$ a quantity. We say that Q is recoverable from Π if it factors through the actual image $\Pi(X)$.

Proposition 6.2 (Correct fiber criterion). *There exists a unique map*

$$\hat{Q} : \Pi(X) \longrightarrow Z \quad \text{such that} \quad Q = \hat{Q} \circ \Pi$$

if and only if

$$\Pi(x) = \Pi(x') \implies Q(x) = Q(x') \quad (x, x' \in X).$$

If Π is surjective, then $\Pi(X) = Y$. Without surjectivity, the criterion does not determine values of an extension $Y \rightarrow Z$ outside $\Pi(X)$.

Proof. Factorization implies fiber constancy. Conversely, define $\hat{Q}(\Pi(x)) = Q(x)$. Fiber constancy makes the definition independent of the representative and uniqueness holds on $\Pi(X)$. $\square$

Corollary 6.3 (Continuous descent). *Let X, Y, Z be topological spaces and let $\Pi : X \rightarrow Y$ be a surjective quotient map. A continuous map $Q : X \rightarrow Z$ admits a unique continuous factorization $Q = \hat{Q} \circ \Pi$ if and only if it is constant on the fibers of Π .*

Proof. Apply Proposition 6.2. The defining property of the quotient topology says that $\hat{Q}$ is continuous exactly when $\hat{Q} \circ \Pi = Q$ is continuous. $\square$

Remark 6.4. If smooth, measurable, Lipschitz, or other regular descent is required, the corresponding category needs its own hypotheses. Set-theoretic fiber constancy alone proves only set-theoretic factorization.

6.2 Compact-group symmetry

Definition 6.5 (Equivariant spectral datum). Let G be a compact group with a strongly continuous unitary representation $U : G \rightarrow \mathcal{U}(\mathcal{H})$. The datum $(\mathcal{H}, L, U)$ is equivariant if

$$U_g \text{Dom}(L) = \text{Dom}(L), \quad LU_g x = U_g Lx \quad (x \in \text{Dom}(L)).$$

Proposition 6.6 (Static symmetry labels). *If L has compact resolvent and $(\mathcal{H}, L, U)$ is equivariant, then every spectral projection $\mathbf{1}_W(L)$ commutes with U_g . In particular, each finite-dimensional eigenspace*

$$E_\lambda = \ker(L - \lambda I)$$

is a unitary G -representation and decomposes into irreducibles.

Proof. The domain invariance and equality $LU_g = U_g L$ imply

$$U_g(L + I)^{-1} = (L + I)^{-1}U_g.$$

The bounded functional calculus therefore gives commutation with every bounded Borel function of L , hence with each spectral projection. The eigenspaces are finite-dimensional by compact resolvent. Their restricted G -actions are unitary, so invariant subspaces have invariant orthogonal complements; iteration gives a finite direct sum of irreducibles. $\square$

Warning 6.7. Proposition 6.6 is a static statement. It does not assert that the multiplicity of a fixed eigenvalue remains constant under perturbation. Eigenvalues may split, merge, or cross. Quantitative stability of an isolated cluster requires a declared spectral gap and control of the corresponding Riesz projection; that problem is not treated here.

6.3 A corrected toric internal model

Let V be a finite-dimensional unitary G -representation and consider

$$\mathcal{H} = L^2(\mathbb{T}^d) \otimes V, \quad L = L_A \otimes \mathbf{I}_V.$$

Here G acts trivially on $L^2(\mathbb{T}^d)$ and through the given representation on V .

Proposition 6.8 (Full eigenspace at a toric level). *For every eigenvalue λ of L_A ,*

$$E_\lambda(L) = \bigoplus_{\substack{n \in \mathbb{Z}^d \\ n^\top A n = \lambda}} \mathbb{C} e_n \otimes V.$$

Consequently, as a G -representation,

$$E_\lambda(L) \cong V^{\oplus r_A(\lambda)}.$$

If $H \subset G$ is closed and

$$V \downarrow_H \cong \bigoplus_{\alpha} b_{\alpha} W_{\alpha},$$

then

$$E_\lambda(L) \downarrow_H \cong \bigoplus_{\alpha} (r_A(\lambda) b_{\alpha}) W_{\alpha}.$$

Proof. Proposition 5.2 identifies all Fourier modes with eigenvalue λ . Tensoring each Fourier line with V gives the full eigenspace. The group acts trivially on the Fourier multiplicity space and through the given representation on V , so multiplicities multiply by $r_A(\lambda)$. $\square$

Remark 6.9. This proposition supplies representation-theoretic labels attached to a spectral level. It does not identify those labels with physical fields, charges, masses, or interactions.

7 Claim boundary and admissibility test

The framework is intentionally narrow. The following table records the main logical boundaries.

Invalid shortcut	Missing hypothesis or conclusion
Compact carrier $\Rightarrow$ discrete spectrum	False by Example 2.4; an operator theorem such as ellipticity and compact Sobolev embedding is required.
Compact resolvent $\Rightarrow$ heat trace for every $t > 0$	False by Example 2.8; trace-class summability is separate.
Formal q -series $\Rightarrow$ spectrum	Levels, integer multiplicities, convergence, and an operator realization are missing.
Abstract diagonal realization $\Rightarrow$ compact geometry	False in general; Weyl and heat asymptotics impose geometric obstructions, as in Example 4.7.
$\phi \in \mathbb{R}/2\pi\mathbb{Z}$ for arbitrary real levels	False unless the spectrum satisfies the arithmetic condition in Proposition 3.4.
Fiber constancy $\Rightarrow$ a map on all of Y	It canonically produces a map only on $\Pi(X)$; surjectivity or a separate extension is needed.
Equivariant eigenspace $\Rightarrow$ perturbative label stability	False without an isolated cluster, a gap, and an equivariant perturbation theorem.
Representation branch $\Rightarrow$ physical sector	Dynamics, units, observables, and empirical identification remain absent.

7.1 Audit protocol

For use as a diagnostic rather than as a theorem, a proposed spectral encoding should state, in order:

1. the index set and level function;
2. the meaning and sign/integrality of the coefficients;
3. the convergence domain;
4. the operator and its self-adjoint domain, if an operator claim is made;
5. compact-resolvent and heat-admissibility hypotheses separately;
6. the carrier, bundle, locality, and ellipticity data, if a geometric claim is made;
7. the group action and commutation relations, if symmetry labels are used;
8. the readout map and the category in which descent is asserted;
9. any additional bridge data needed for a physical interpretation.

Items 1–3 define the analytic expression. Items 4–5 justify abstract spectral and heat language. Item 6 is needed to pass from an abstract operator to geometry. Items 7–8 type symmetry and recoverability statements. Item 9 is logically external to the mathematical spectral package.

8 Conclusion

The correct foundational implication is

$$L = L^* \geq 0, \quad (L + I)^{-1} \text{ compact} \implies \text{discrete finite-multiplicity spectrum,}$$

with

$$e^{-tL} \in \mathcal{S}_1 \text{ for every } t > 0 \implies Z_L(z) = \int e^{-z\lambda} d\mu_L(\lambda), \quad \Re z > 0.$$

Compactness of a carrier is not a substitute for the first hypothesis, and compact resolvent is not a substitute for the second. Closed Laplace-type geometry supplies both and adds Weyl and local heat asymptotics. Abstract diagonal realizability supplies neither locality nor geometry.

The complex-time formulation removes the ambiguous use of a periodic phase: periodicity is a theorem only when the spectrum lies in the corresponding arithmetic lattice. The corrected readout criterion likewise produces a canonical factor only on the actual image unless surjectivity is imposed. Finally, equivariance gives representation structure on eigenspaces but not perturbative stability or physical interpretation.

These distinctions make the framework suitable as an expository companion to more focused research. Natural research questions – geometric realizability obstructions, quantitative equivariant cluster stability, and branch-resolved heat-trace inversion – require separate theorems and are intentionally not claimed here.

References

- [1] N. Berline, E. Getzler, and M. Vergne, *Heat Kernels and Dirac Operators*, Grundlehren der mathematischen Wissenschaften 298, Springer-Verlag, Berlin, 1992. doi:10.1007/978-3-642-58088-8.
- [2] A. H. Chamseddine and A. Connes, The spectral action principle, *Communications in Mathematical Physics* **186** (1997), 731–750. doi:10.1007/s002200050126.
- [3] I. Chavel, *Eigenvalues in Riemannian Geometry*, Pure and Applied Mathematics 115, Academic Press, Orlando, 1984.

- [4] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.
- [5] E. B. Davies, *Heat Kernels and Spectral Theory*, Cambridge Tracts in Mathematics 92, Cambridge University Press, Cambridge, 1989. doi:10.1017/CBO9780511566158.
- [6] G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed., CRC Press, Boca Raton, 2016. doi:10.1201/b19172.
- [7] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, New York, 1991. doi:10.1007/978-1-4612-0979-9.
- [8] P. B. Gilkey, The spectral geometry of a Riemannian manifold, *Journal of Differential Geometry* **10** (1975), 601–618. doi:10.4310/jdg/1214433164.
- [9] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer-Verlag, Berlin, 1976; reprinted in Classics in Mathematics, Springer, 1995. doi:10.1007/978-3-642-66282-9.
- [10] S. Minakshisundaram and Å. Pleijel, Some properties of the eigenfunctions of the Laplace-operator on Riemannian manifolds, *Canadian Journal of Mathematics* **1** (1949), 242–256. doi:10.4153/CJM-1949-021-5.
- [11] D. Mumford, *Tata Lectures on Theta I*, Progress in Mathematics 28, Birkhäuser, Boston, 1983.
- [12] M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, rev. ed., Academic Press, San Diego, 1980.
- [13] M. Reed and B. Simon, *Methods of Modern Mathematical Physics IV: Analysis of Operators*, Academic Press, New York, 1978.
- [14] R. T. Seeley, Complex powers of an elliptic operator, in *Singular Integrals*, Proceedings of Symposia in Pure Mathematics 10, American Mathematical Society, Providence, 1967, pp. 288–307, MR 0237943.
- [15] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., Mathematical Surveys and Monographs 120, American Mathematical Society, Providence, 2005. doi:10.1090/surv/120.
- [16] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, Princeton, 2003.